

E5: Text Embeddings by Weakly-Supervised Contrastive Pre-training

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Introduction

- Pre-trained LM (BERT, GPT) -> single-vector embedding을 요하는 작업에 이상적이지 않다.
(효율성, 다용성 면에서)
- Contrastive Learning (대조 학습)
: sequence-level representations를 향상시키는 대표적 방식 중 하나.
 - GTR, Sentence-T5 : passage retrieval, STS tasks에서의 임베딩 학습
 - But, Task-specific supervised datasets으로 fine-tuning => 범용적이지 않다.
 - Unsupervised embeddings from automatically constructed text pairs
 - 양적으로는 충분했으나, 좋은 퀄리티를 가지지 못했다는 단점.



Introduction

- E5 : EmbEddings from bidirEctional Encoder rEpresentations.
 - 네이밍 참..억까
 - 직역하자면 양방향 인코더 표현 기반 임베딩
 - E5 aims to provide strong off-the-shelf text embeddings **suitable for any tasks** requiring single-vector representations in both zero-shot or fine-tuned settings.
 - low-quality synthetic text pairs 대신, 자체적으로 CCPairs datasets을 통해 contrastive train.
 - 2022년 논문 발표 당시 기준, 1B 미만의 적은 파라미터를 가지고 MTEB에서 SOTA 달성
 - 비슷한 성능을 내던 GTR, Sentence-T5는 E5보다 40배 많은 파라미터 보유
- => **self-supervised pre-training만으로도 고품질 임베딩 학습이 가능하다는 점을 보여준 점에 의의**



CCPairs (Colossal Clean text Pairs)

- **Large Collection of Text Pairs Dataset** from web sources

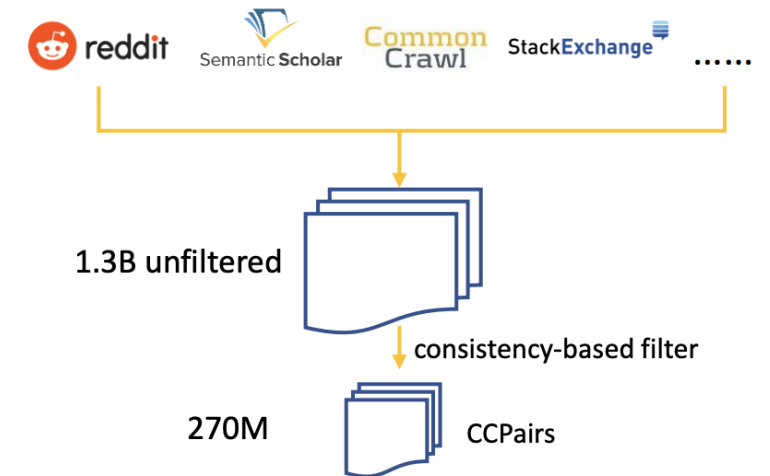
기존 방식

small-scale human-annotated data 활용 or Heuristic을 통해 large-scale but **very noisy supervision** signals 수집.



CCPairs

먼저 semi-structured data 수집 후 학습 (called Noisy Large Dataset), 그 후 **Consistency-based filtering**을 통해 **high-quality data** 선별



CCPairs (Colossal Clean text Pairs)

1. Harvesting semi-structured data sources

(Query, Passage) Text Pair Dataset을 Web으로부터 수집한다.

Passage meaning in this paper : short sentence, paragraph, long document, ...

data source	type of text pairs	random example	# of pairs
Wikipedia	(entity+section title, passage)	q: Lexden History p: The site on which Lexden now stands was crossed by the fortifications of iron age Colchester...	24M
Reddit	(post, upvoted comment)	q: What makes a client good quality to you? I'm putting together my ideal client ... p: Respectful of schedules. And pays on time....	60M
Common Crawl	(title, passage)	q: Central Intake Unit Broome County p: Caseworkers from Central Intake assess the household and risk of placement. If eligible...	69M
Stackexchange	(title, answer) (title+description, answer)	q: Will killing Python made problems for Apache p: Python and Apache aren't related, unless your app is making use of Python. ...	19M
S2ORC	(title, abstract) (title, citation title) (abstract, citation abstract)	q: Constructive Dual DP for Reservoir Optimization p: Dynamic programming (DP) is a well established technique for optimization of reservoir manage...	90M
News	(title, passage) (highlight, passage)	q: LG Display reports Q1 operating loss as... p: April 25 (Reuters) - South Korea's LG Display Co Ltd reported its first quarterly operating loss...	3M
Others	misc.	misc.	6M
All above	-	-	~ 270M

-> Collected **1.3B** noisy text pairs
(대부분 Reddit and Common Crawl)



CCPairs (Colossal Clean text Pairs)

2. Consistency-based filter

먼저 1.3B Noisy Text Pairs로 모델을 학습한 후, 필터링 진행

[Consistency-based Filtering]

1. 각 Query에 대해, " **정답 Passage + 1 million random passages**" 후보군을 랭킹
2. **Top-k 안에 원래 정답이 존재하는 text pairs만 잔류**, 나머지는 필터링 진행. (논문에서는 **k=2**로 진행)

=> 270M high-quality CCPairs 수집

The **intuition** for this technique comes from **the memorization behaviors of neural networks**: when trained on noisy datasets, neural networks **tend to memorize the clean labels first** and then **gradually overfit the noisy labels**. Similar techniques have been widely used for removing dataset noises.

=> Maybe, Noisy Text Pairs 모델 학습도 **적은 steps** 동안만 학습하지 않았을까 생각합니다.



Method

First-stage : contrastive pre-training with unlabeled text pairs from CCPairs.

Second-stage : fine-tuning on small, high-quality labeled datasets



Method

First-stage : contrastive pre-training with unlabeled text pairs from CCPairs.

Contrastive pre-training : relevant text pairs를 irrelevant / negative text pairs와 구별함에 목적.

- **InfoNCE** 손실함수를 이용한 in-batch negatives contrastive learning

in-batch negatives contrastive learning이란?

: 같은 mini-batch 안에 있는 다른 샘플들을 **negative**로 간주하는 방식.

Contrastive learning에는 Positive pair(의미적으로 유사한 문장 쌍)과 Negative pair(의미적으로 다른 문장 쌍)이 필요한데, 직접 negative. pair를 수집하는 건 시간이 많이 든다.

=> batch 안의 다른 샘플들을 자동으로 negative로 활용하는 방식

$$\min L_{\text{cont}} = -\frac{1}{n} \sum_i \log \frac{\overset{\text{Positive pair}}{e^{s_{\theta}(q_i, p_i)}}}{\underset{\text{Positive pair}}{e^{s_{\theta}(q_i, p_i)}} + \sum_j \underset{\text{Negative pair}}{e^{s_{\theta}(q_i, p_{ij}^-)}}}$$

Batch size : 32,768

Score function : cosine similarity / temperature (0.01)

$$s_{\theta}(p, q) = \cos(\mathbf{E}_q, \mathbf{E}_p) / \tau$$



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Q. 다른 걸 몽땅 **negative**로 잡는 게 효과가 있나?

A. **Batch size**가 꽤 크다는 전제에선 먹히더라. 오히려 더 안정적이고 좋던데?(MoCo)

Batch size : **32,768**

Score function : cosine similarity / temperature (0.01)

$$s_{\theta}(p, q) = \cos(\mathbf{E}_q, \mathbf{E}_p) / \tau$$



Method

First-stage : contrastive pre-training with unlabeled text pairs from CCPairs.

Shared encoder

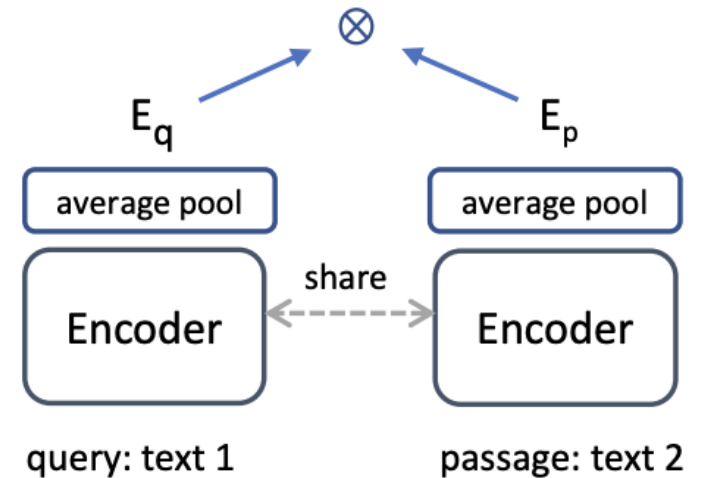
Query와 passage를 공유 인코더를 통해 표현.

→ Q, P가 같은 임베딩 공간에서 표현되므로 직접적인 의미 비교 가능

Q, P 간 차이점만 주기 위해 prefix만 붙여주었음.

Q -> "query: ~~~~"

P -> "passage: ~~~~~"



Method

Second-stage : fine-tuning on small, high-quality labeled datasets

- further training on labeled data can inject human knowledge into the model to boost the performance.

Datasets

- | | | |
|---|---|--------------------|
| 1. NLI (Natural Language Inference) dataset | → | For STS task |
| 2. MS-MARCO passage ranking dataset | } | For Retrieval task |
| 3. NQ (Natural Questions) dataset | | |

Selecting Hard Negatives

MS-MARCO, NQ (STS) : **mined hard negatives and knowledge distillation** from a cross-encoder (CE) teacher model

NLI (Retrieval) : **contradiction sentences** are regarded as hard negatives.



Method

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Loss function : linear interpolation

(contrastive loss for hard labels + KL divergence for distilling soft labels from teacher model)

$$\min D_{\text{KL}}(p_{\text{ce}}, p_{\text{stu}}) + \alpha L_{\text{cont}}$$

P_{ce}, P_{stu} : Probability from the cross-encoder teacher/student model

[Fine-tuning 과정에 대해 자세히 정리한 블로그 참고](#)



Evaluation

BEIR Benchmark (Information Retrieval datasets collection)

Use 15 datasets of whole 19 IR dataset in this evaluation.

Metric: nDCG@10

MTEB Benchmark (Massive Text Embedding Benchmark)

Classification, Clustering, Pair Classification, Rerank, Retrieval, STS, Summarization datasets.

Metric : accuracy, v-measure, average precision, MAP, nDCG@10, Spearman coefficients



Results - BEIR

Results with Unsupervised Methods

Table 1: Unsupervised methods on the BEIR benchmark (nDCG@10). For SimCSE, we report results with BERT_{base}. cpt_{300M} [41] is only available through paid API and evaluation results on some datasets are missing in the original paper. The highest number for each dataset is in bold, and the second highest is underlined. †: we report the LaPraDor [62] results without ensembling with BM25. *: reproduction with the released checkpoint.

	BM25	SimCSE	LaPraDor [†]	Contriever	cpt _{300M}	E5-PT _{small}	E5-PT _{base}	E5-PT _{large}
MS MARCO	22.8	9.4	16.9*	20.6	19.9	25.4	<u>26.0</u>	26.2
Trec-Covid	65.6	26.2	22.7	27.4	52.9	52.0	61.0	<u>61.8</u>
NFCorpus	32.5	9.9	31.1	31.7	32.0	29.3	35.8	<u>33.7</u>
NQ	32.9	11.7	18.1	25.4	-	37.3	<u>39.0</u>	41.7
HotpotQA	60.3	19.8	30.3	48.1	51.5	46.0	<u>52.4</u>	52.2
FiQA	23.6	9.8	20.3	24.5	34.1	38.3	<u>40.0</u>	43.2
ArguAna	31.5	38.3	45.9	37.9	38.7	42.5	42.2	<u>44.4</u>
Touche-2020	36.7	8.9	9.4	19.3	<u>21.0</u>	19.9	16.9	19.8
CQADupStack	29.9	13.2	22.0	28.4	-	35.0	<u>35.4</u>	38.9
Quora	78.9	78.0	78.7	83.5	68.1	<u>85.8</u>	85.7	86.1
DBPedia	31.3	15.0	25.0	29.2	27.2	34.5	<u>35.4</u>	37.1
Scidocs	15.8	5.5	13.3	14.9	-	19.9	<u>21.1</u>	21.8
Fever	75.3	21.1	36.8	68.2	57.1	62.5	63.4	68.6
Climate-Fever	21.3	11.8	13.8	15.5	<u>15.8</u>	14.5	15.4	15.7
Scifact	66.5	25.7	55.5	64.9	65.4	68.5	73.7	72.3
Average	41.7	20.3	29.3	36.0	-	40.8	<u>42.9</u>	44.2
Best on	2	0	1	0	0	0	2	7

First reported result that
an **unsupervised model can beat BM25 on the BEIR.**



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Average	41.7	20.3	29.3	36.0	-	40.8	<u>42.9</u>	44.2
Best on	<u>5</u>	0	1	0	0	0	2	7

Quantitative dataset

Contriever : adopts random cropping

LaPraDor : ICT + dropout-as-positive-instance from SimCSE

VS

E5 : more effort in dataset curation (For Qualitative dataset)



Results - BEIR

Results with Supervised Fine-tuning

Table 2: Supervised fine-tuning results on the BEIR benchmark. Results for ANCE [61], ColBERT [31] and Contriever come from Izacard et al. [28]. The best result is in bold, and the second best is underlined.

	ANCE	GTR _{base}	ColBERT	Contriever	cpt _{300M}	GTR _{large}	E5 _{small}	E5 _{base}	E5 _{large}
MS MARCO	38.8	42.0	40.1	40.7	-	43.0	42.3	<u>43.1</u>	44.1
Trec-Covid	65.4	53.9	67.7	59.6	67.9	55.7	76.8	79.6	<u>78.3</u>
NFCorpus	23.7	30.8	30.5	32.8	33.2	32.9	33.9	36.6	<u>36.1</u>
NQ	44.6	49.5	52.4	49.8	-	54.7	58.7	<u>60.0</u>	62.9
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Fever	66.9	66.0	77.1	<u>75.8</u>	72.1	71.2	53.5	58.2	65.0
Climate-Fever	19.8	<u>24.1</u>	18.4	23.7	18.5	26.2	15.8	15.4	22.4
Scifact	50.7	60.0	67.1	67.7	67.2	63.9	65.6	73.1	72.6
Average	40.5	44.0	44.4	46.6	-	47.0	<u>46.0</u>	<u>48.7</u>	50.0
Best on	0	0	1	1	1	<u>4</u>	0	3	5



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NFCorpus	23.7	30.8	30.5	32.8	33.2	32.9	33.9	36.6	<u>36.1</u>
NQ	44.6	49.5	52.4	49.8	-	54.7	58.7	<u>60.0</u>	62.9
HotpotQA	45.6	53.5	59.3	63.8	59.4	<u>57.9</u>	56.3	<u>62.2</u>	<u>63.3</u>
FiQA	29.5	34.9	31.7	32.9	38.4	42.4	34.8	36.4	<u>38.6</u>
ArguAna	41.5	51.1	23.3	44.6	47.0	52.5	46.7	<u>51.4</u>	49.4
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Scifact	50.7	60.0	67.1	67.7	67.2	63.9	65.6	73.1	<u>72.6</u>
Average	40.5	44.0	44.4	46.6	-	47.0	46.0	<u>48.7</u>	50.0
Best on	0	0	1	1	1	<u>4</u>	0	3	5

Due to the lack of enough domain diversity for the fine-tuning datasets



Results - MTEB

Table 3: Results on the MTEB benchmark [40] (56 datasets in English subset). Here we only report averaged numbers on each task category for space reasons, please check out Appendix B for a detailed version. BERT-FT_{base} uses the same fine-tuning data as E5 but initializes from BERT_{base}.

# of datasets →	Class. 12	Clust. 11	PairClass. 3	Rerank 4	Retr. 15	STS 10	Summ. 1	Avg 56
<i>Unsupervised models</i>								
Glove	57.3	27.7	70.9	43.3	21.6	61.9	28.9	42.0
BERT	61.7	30.1	56.3	43.4	10.6	54.4	29.8	38.3
SimCSE-BERT-unsup	62.5	29.0	70.3	46.5	20.3	74.3	31.2	45.5
E5-PT _{small}	67.0	41.7	78.2	53.1	40.8	68.8	32.7	54.3
E5-PT _{base}	67.9	43.4	79.2	53.5	42.9	69.5	31.1	55.6
E5-PT _{large}	69.0	44.3	80.3	54.4	44.2	69.9	32.6	56.6
<i>Supervised models</i>								
SimCSE-BERT-sup	67.3	33.4	73.7	47.5	21.8	79.1	23.3	48.7
BERT-FT _{base}	68.7	33.9	82.6	50.5	41.5	79.2	29.0	55.2
Contriever	66.7	41.1	82.5	53.1	41.9	76.5	30.4	56.0
GTR _{large}	67.1	41.6	85.3	55.4	47.4	78.2	29.5	58.3
Sentence-T5 _{large}	72.3	41.7	85.0	54.0	36.7	81.8	29.6	57.1
E5 _{small}	71.7	39.5	85.1	54.5	46.0	80.9	31.4	58.9
E5 _{base}	72.6	42.1	85.1	55.7	48.7	81.0	31.0	60.4
E5 _{large} 300M	73.1	43.3	85.9	56.5	50.0	82.1	31.0	61.4
<i>Larger models</i>								
GTR _{xxl}	67.4	42.4	86.1	56.7	48.5	78.4	30.6	59.0
Sentence-T5 _{xxl} 4.8B	73.4	43.7	85.1	56.4	42.2	82.6	30.1	59.5

SOTA

SOTA



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BERT	61.7	30.1	56.3	43.4	10.6	54.4	29.8	38.3
SimCSE-BERT-unsup	62.5	29.0	70.3	46.5	20.3	74.3	31.2	45.5
E5-PT _{small}	67.0	41.7	78.2	53.1	40.8	68.8	32.7	54.3
E5-PT _{base}	67.9	<u>43.4</u>	79.2	53.5	42.9	69.5	31.1	55.6
E5-PT _{large}	69.0	44.3	80.3	54.4	44.2	69.9	<u>32.6</u>	56.6
<i>Supervised models</i>								
SimCSE-BERT-sup	67.3	33.4	73.7	47.5	21.8	79.1	23.3	48.7
BERT-FT _{base}	68.7	33.9	82.6	50.5	41.5	79.2	29.0	55.2
Contriever	66.7	41.1	82.5	53.1	41.9	76.5	30.4	56.0
GTR _{large}	67.1	41.6	<u>85.3</u>	55.4	47.4	78.2	29.5	58.3
Sentence-T5 _{large}	72.3	41.7	<u>85.0</u>	54.0	36.7	<u>81.8</u>	29.6	57.1
E5 _{small}	71.7	39.5	85.1	54.5	46.0	80.9	31.4	58.9
E5 _{base}	<u>72.6</u>	42.1	85.1	<u>55.7</u>	<u>48.7</u>	81.0	31.0	<u>60.4</u>
E5 _{large}	73.1	43.3	85.9	56.5	50.0	82.1	31.0	61.4
<i>Larger models</i>								
GTR _{xxl}	67.4	42.4	86.1	56.7	48.5	78.4	30.6	59.0
Sentence-T5 _{xxl}	73.4	43.7	85.1	56.4	42.2	82.6	30.1	59.5

Usefulness of contrastive pre-training on CCPairs



Results - MTEB

Table 3: Results on the MTEB benchmark [40] (56 datasets in English subset). Here we only report averaged numbers on each task category for space reasons, please check out Appendix B for a detailed version. BERT-FT_{base} uses the same fine-tuning data as E5 but initializes from BERT_{base}.

# of datasets →	Class. 12	Clust. 11	PairClass. 3	Rerank 4	Retr. 15	STS 10	Summ. 1	Avg 56
<i>Unsupervised models</i>								
Glove	57.3	27.7	70.9	43.3	21.6	61.9	28.9	42.0
BERT	61.7	30.1	56.3	43.4	10.6	54.4	29.8	38.3
SimCSE-BERT-unsup	62.5	29.0	70.3	46.5	20.3	74.3	31.2	45.5
E5-PT _{small}	67.0	41.7	78.2	53.1	40.8	68.8	32.7	54.3
E5-PT _{base}	67.9	<u>43.4</u>	79.2	53.5	42.9	69.5	31.1	55.6
E5-PT _{large}	69.0	44.3	80.3	54.4	44.2	69.9	<u>32.6</u>	56.6
<i>Supervised models</i>								
SimCSE-BERT-sup	67.3	33.4	73.7	47.5	21.8	79.1	23.3	48.7
BERT-FT _{base}	68.7	33.9	82.6	50.5	41.5	79.2	29.0	55.2
Contriever	66.7	41.1	82.5	53.1	41.9	76.5	<u>30.4</u>	56.0
GTR _{large}	67.1	41.6	<u>85.3</u>	55.4	47.4	78.2	29.5	58.3
Sentence-T5 _{large}	72.3	41.7	<u>85.0</u>	54.0	36.7	81.8	29.6	57.1
E5 _{small}	71.7	39.5	85.1	54.5	46.0	80.9	31.4	58.9
E5 _{base}	<u>72.6</u>	42.1	85.1	<u>55.7</u>	<u>48.7</u>	81.0	31.0	<u>60.4</u>
E5 _{large}	73.1	43.3	85.9	56.5	50.0	82.1	31.0	61.4
<i>Larger models</i>								
GTR _{xxl}	67.4	42.4	86.1	56.7	48.5	78.4	30.6	59.0
Sentence-T5 _{xxl}	73.4	43.7	85.1	56.4	42.2	82.6	30.1	59.5

Performance improves after SFT



Analysis

Impacts of Batch Size

: larger batch size -> provide more negatives -> improve the quality of learned text embeddings

Table 5: Impacts of different batch sizes for contrastive pre-training.

batch size	NFCorpus	NQ	FiQA	Quora	DBPedia	Scifact	Avg
32k	35.8	39.0	40.0	85.7	35.4	73.7	51.6
8k	33.3	38.5	37.6	85.7	34.0	71.8	50.2
1k	28.2	33.1	30.4	84.0	30.1	69.1	45.8

Fine-tuning Datasets

: importance of dataset diversity for learning text embeddings

Table 6: Fine-tuning with different combinations of labeled data.

Fine-tuned on	Retrieval	STS	Classification	Summ.	MTEB Avg
No fine-tuning	42.9	69.5	67.9	31.1	55.6
MS-MARCO + NQ	50.3	78.3	68.3	30.6	59.0
NLI	38.3	81.1	72.6	31.6	57.3
All above	48.7	81.0	73.1	31.0	60.4



Analysis

Data Filtering

: CCPairs 에서 사용한 Consistency-based filter가 실제로 quality enhancement에 효과적이었음.

Table 7: Data filtering. For the top 2 rows, we train with 1M random text pairs.

# of pairs		NFCorpus	NQ	FiQA	Quora	DBPedia	Scifact	Avg
1M	w/o filter	23.0	15.1	18.5	83.1	18.2	51.4	34.9
	w/ filter	26.8	22.7	24.5	85.0	27.5	57.5	40.7
All	w/o filter	34.5	35.4	39.1	85.7	32.9	72.5	50.0
	w/ filter	35.8	39.0	40.0	85.7	35.4	73.7	51.6

* w/o filter is **4x more data** than w/ filter in All of pairs

BM25 VS Dense Retrieval

: 아직은 BM25가 더 명확한 이점을 취한다. (simplicity, efficiency, interpretability, ..etc)



Conclusion

- E5 : strong off-the-shelf performance for a **wide range of tasks** (Retrieval, STS, Text Matching..)
- E5 achieves superior fine-tuned performance than existing MTEB benchmark datasets which have 40x more parameters



Future works

- **E5-Mistral-7B**

: E5에서의 한계를 (Dataset 구축에 Engineering Effort 많이 듦) 해결하기 위해 Text-Pair 데이터셋 생성과 임베딩 추출을 LLM이 생성.

- **E5-Instruct**

: E5-Mistral-7B에서 LLM으로 Text-Pair 데이터 증강하는게 좋다는 걸 확인한 후, LLM 실사용 시 드는 비용을 줄이기 위해 BERT-style encoder에 학습시킨 모델.
E5-Mistral-7B에 거의 준하는 성능



References.

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